

Ex A Contractor purchases a shipment of 100 transistors. It is his policy to test of these transistors and to keep the shipment only if at least 9 of the 10 are in working condition. If the shipment contains 20 defective transistors, what is the probability it will be kept.

Sol $p = \frac{80}{100} = 0.8, \quad q = \frac{20}{100} = 0.2$

$$\begin{aligned} P(X=9) + P(X=10) &= \text{the required prob.} \\ &= \binom{10}{9} (0.8)^9 (0.2)^1 + \binom{10}{10} (0.8)^{10} (0.2)^0 \\ &= 0.2684 + 0.1073 = 0.3757 \end{aligned}$$

Ex. In a precision bombing attack there is a 50% chance that any one bomb will strike the target. Two direct hit are required to destroy the target completely. How many bombs must be dropped to give a 99% chance or better of completely destroying the target?

Sol. We have:

p = Probability that the bomb strikes the target.
 Let n be the number of bombs which should be dropped to ensure 99% chance or better of completely destroying the target. This implies that a prob. that out of n bombs at least two strike the target, is greater than 0.99.

Let X be a r.v. representing the number of bombs striking the target. Then $X \sim B(n, p)$ with

$$f(x) = P(X=x) = \binom{n}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{n-x} \\ = \binom{n}{x} \left(\frac{1}{2}\right)^n; \quad x = 0, 1, 2, \dots, n.$$

We should have:

$$P(X \geq 2) \geq 0.99$$

$$\left[1 - P(X \leq 1) \right] \geq 0.99$$

$$\Rightarrow \left[1 - \{P(0) + P(1)\} \right] \geq 0.99$$

$$\Rightarrow 1 - \left\{ \binom{n}{0} + \binom{n}{1} \right\} \left(\frac{1}{2}\right)^n \geq 0.99$$

Appointments

1. During war 1 ship out of 9 was sunk on an average in making a certain voyage. What is the probability that exactly 3 out of a Convoy of 6 ships would arrive. Ans. $6C_3 \left(\frac{8}{9}\right)^3 \left(\frac{1}{9}\right)^{6-3}$
 $p = \frac{8}{9}, q = \frac{1}{9}$